

DERIVATION OF DC-LINK CAPACITOR AND DOUBLE FREQUENCY RIPPLE POWER

The DC bus voltage is composed of a steady-state component V_0 and a second-harmonic (double-frequency) ripple term, given as:

$$V_g^{\text{DC}} = V_g^{\text{DC},0} + \underbrace{V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}})}_{\text{double frequency ripple}} \quad (1)$$

Where V_g^{DC} is the instantaneous DC bus voltage of inverter g , $V_g^{\text{DC},0}$ is the steady state component of the DC bus voltage, $V_g^{\text{DC},2\omega}$ is the amplitude of the double line-frequency ripple, ω is the fundamental grid angular frequency, and θ_g^{DC} is the phase angle of the double-frequency voltage ripple.

Similarly, the DC bus current is

$$I_g^{\text{DC}} = I_g^{\text{DC},0} + I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \quad (2)$$

The phase shift between voltage and current ripple is absorbed into the phase angle θ_g^{DC} for simplicity.

Also, the capacitor current is given by the time derivative of (1),

$$I_g^{\text{DC}} = C_g \frac{dV_g^{\text{DC}}}{dt}. \quad (3)$$

In relation to (1) will yield,

$$\begin{aligned} I_g^{\text{DC}} &= C_g \frac{d}{dt} [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}})] \\ &= 2\omega C_g V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_g^{\text{DC}}). \end{aligned} \quad (4)$$

Hence, the amplitude of the ripple current across the capacitor is

$$I_g^{\text{DC},2\omega} = 2\omega C_g V_g^{\text{DC},2\omega} \quad (5)$$

Where $I_{g,t}^{\text{DC}}$ is the instantaneous DC bus current of inverter g at time t , $I_g^{\text{DC},0}$ is the average (DC) component of the DC bus current, $I_g^{\text{DC},2\omega}$ is the amplitude of the double line-frequency ripple current, ω is the fundamental grid angular frequency, and θ_g^{DC} is the phase angle of the double-frequency current ripple.

The double-frequency power injected into the DC bus can be obtained through the instantaneous DC power derived from the product of (1) and (2),

$$\begin{aligned} \mathbf{P}_g^{\text{DC}} &= V_g^{\text{DC}} I_g^{\text{DC}} \\ &= \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},0}}_{\text{average power}} + \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_g^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},2\omega} \sin^2(2\omega t + \theta_g^{\text{DC}})}_{\text{higher-order term}} \end{aligned} \quad (6)$$

Focusing on the double-frequency contributions, and isolating the dominant 2ω component through harmonic decomposition of the DC voltage and current (while neglecting higher-order coupling terms), we have:

$$\begin{aligned} \mathbf{P}_g^{\text{DC},2\omega} &= V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \\ &\quad + V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_g^{\text{DC}}) \end{aligned} \quad (7)$$

From (2),

$$I_g^{\text{DC},0} = I_g^{\text{DC}} - I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \quad (8)$$

Substituting (8) into the (7) yields:

$$\begin{aligned} \mathbf{P}_g^{\text{DC},2\omega} &= V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \\ &\quad + V_g^{\text{DC},2\omega} (I_g^{\text{DC}} - I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}})) \\ &\quad \times \sin(2\omega t + \theta_g^{\text{DC}}). \end{aligned} \quad (9)$$

Expanding the expression gives:

$$\begin{aligned} \mathbf{P}_g^{\text{DC},2\omega} &= V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \\ &\quad + V_g^{\text{DC},2\omega} I_g^{\text{DC}} \sin(2\omega t + \theta_g^{\text{DC}}) \\ &\quad - V_g^{\text{DC},2\omega} I_g^{\text{DC},2\omega} \sin^2(2\omega t + \theta_g^{\text{DC}}). \end{aligned} \quad (10)$$

Substituting (4) and (5) into (10) and apply the trigonometric identity $\frac{1}{2} \sin 2(\alpha) = \sin(\alpha) \cos(\alpha)$, as appropriate:

$$\begin{aligned} \mathbf{P}_g^{\text{DC},2\omega} &= 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \\ &\quad + \omega C_g (V_g^{\text{DC},2\omega})^2 \sin(4\omega t + 2\theta_g^{\text{DC}}) \\ &\quad - 2\omega C_g (V_g^{\text{DC},2\omega})^2 \sin^2(2\omega t + \theta_g^{\text{DC}}). \end{aligned} \quad (11)$$

Since $V_g^{\text{DC},2\omega} \ll V_g^{\text{DC},0}$, all higher-order ripple terms of order $\mathcal{O}((V_g^{\text{DC},2\omega})^2)$ are neglected in the harmonic decomposition, and only the dominant 2ω component is retained.

Hence, the double frequency power is given as:

$$\mathbf{P}_g^{\text{DC},2\omega} = 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_g^{\text{DC}}) \quad (12)$$

The amplitude of this double-frequency component, as given in (13), represents the primary contribution to the DC bus power, which directly affects converter operation.

$$\mathbf{P}_g^{2\omega} = 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \quad (13)$$

DC-LINK CAPACITOR AND RIPPLE POWER COST MODEL

The DC-link capacitor cost follows the analytical expression in [1], where the unit cost of metallized polypropylene film capacitors is a linear function of the rated voltage V_0 and capacitance C_g :

$$C_{\text{dc}} = a_{\text{film}} + b_{\text{film}} V_0 + c_{\text{film}} C_g, \quad (14)$$

with $a_{\text{film}} = -1.19574\$$, $b_{\text{film}} = 0.06429852 \$/\text{V}$, and $c_{\text{film}} = 0.00283842 \$/\mu\text{F}$. Using $V_0 = 1100 \text{ V}$ and $C_g = 40 \mu\text{F}$, the capacitor cost is $C_{\text{dc}} \approx \$69.72$. From (14), the average cost per unit capacitance will be:

$$C_{c_g} = \frac{C_{\text{dc}}}{C_g} = \frac{69.72}{40} \approx 1.743 \$/\mu\text{F} \approx 1.743 \times 10^6 \$/F. \quad (15)$$

A. Relationship between the Capacitance and ripple power

From (13), the capacitance C_g is given as:

$$C_g = \frac{\mathbf{P}_g^{2\omega}}{2\omega V_g^{\text{DC},0} V_g^{\text{DC},2\omega}} \quad (16)$$

$V_g^{\text{DC},0}$ and $V_g^{\text{DC},2\omega}$ can be related using (1) while considering peak-to-peak DC-bus voltage given as:

$$\Delta V_g^{DC,pp} = v^{DC,\max} - v^{DC,\min} = 2V_{DC,2\omega}. \quad (17)$$

where $V_g^{DC,pp}$ is maximum when $\sin(2\omega t + \theta^{DC}) = 1$ and minimum when $\sin(2\omega t + \theta^{DC}) = -1$.

According to [2] and several other literature, the maximum peak-to-peak voltage ripple is given as

$$\Delta V^{DC,pp} \leq 10\% V^{DC,0}, \quad (18)$$

Substituting (17) into (18) yields

$$2V^{DC,2\omega} = 0.10 V^{DC,0}, \quad (19)$$

Hence, the relationship between $V_g^{DC,0}$ and $V_g^{DC,2\omega}$ becomes:

$$V^{DC,2\omega} = 0.05 V^{DC,0}. \quad (20)$$

Substituting (20) into (16), makes C_g becomes:

$$C_g = \frac{\mathbf{P}_g^{2\omega}}{0.1 \omega (V_g^{DC,0})^2} \quad (21)$$

B. Ripple Power Cost Model

The ripple power cost is obtained by constructing a linear relationship between the average cost per unit capacitance in (15) and the relationship between the capacitance and ripple power in (21):

$$C_{dc}(P^{2\omega}) = C_{c_g} \cdot C_g(P^{2\omega}) \quad (22)$$

From (15), $C_{c_g} \approx 1.743 \times 10^6$ \$/F, and from (21), we substitute C_g as a function of ripple power.

Hence, the DC-link capacitor cost as a function of ripple power becomes:

$$C_{dc}(P^{2\omega}) = C_{c_g} \cdot \frac{\mathbf{P}_g^{2\omega}}{0.1 \omega (V_g^{DC,0})^2} \quad (23)$$

The ripple power cost is therefore given as:

$$\frac{dC_{dc}}{dP^{2\omega}} = \frac{C_{c_g}}{0.1 \omega (V_g^{DC,0})^2} \approx 38.2\$/kW \quad (24)$$

Where $\omega = 2\pi f$, $f = 60$ Hz, and $V_g^{DC,0} = V_0 = 1100$ V

REFERENCES

- [1] R. Burkart and J. W. Kolar, "Component cost models for multi-objective optimizations of switched-mode power converters," in *2013 IEEE Energy Conversion Congress and Exposition*, pp. 2139–2146, 2013.
- [2] D. Cittanti, F. Stella, E. Vico, C. Liu, J. Shen, G. Xiu, and R. Bojoi, "Analysis, design, and experimental assessment of a high power density ceramic dc-link capacitor for a 800 v 550 kva electric vehicle drive inverter," *IEEE Transactions on Industry Applications*, vol. 59, no. 6, pp. 7078–7091, 2023.